Supplementary Material: Operator-Specific Forgetting in Matrix-Valued Associative Memory

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1 A. Reproducibility manifest

Reference commit: `cec4ffe`. Tag: `lab-radioactivo-v1`. Repository: `https://github.com/01javierescobar/operator-specific-forgetting`. Archived DOI: 10.5281/zenodo.22051928.

The repository contains implementations, execution runners, evaluation probes, audit logs, model checkpoints, frozen claims, and execution transcripts for experimental milestones O01–O05. The primary trained-model gate is O05. Verification tooling:

- `paper/verify_numbers.py`: Recomputes all quantitative values and consistency tolerances directly from frozen JSON artifacts.
- `paper/make_figures.py`: Regenerates all publication figures (PDF and PNG at 300 DPI) from the same frozen artifacts.

2 B. Notation and memory construction

D denotes the key and value dimension ($D \in \{64, 128\}$). n is the number of stored key–value pairs. $c = (n-1)/D$ is the associative crosstalk load excluding the target item. $\mathbf{w}_i \in \mathbb{C}^D$ are fixed codebook keys with unit-modulus entries ($|\mathbf{w}_{i,k}| = 1$, squared Euclidean norm $\|\mathbf{w}_i\|_2^2 = D$). $\mathbf{v}_i \in \mathbb{R}^D$ are real-valued target vectors. The memory state $\mathbf{M} \in \mathbb{C}^{D\times D}$ stores associations via matrix superposition:

$$\mathbf{M} = \sum_{i=1}^{n} \mathbf{w}_i \mathbf{v}_i^\top. \tag{1}$$

The matched-filter query read using key $\mathbf{w}_j$ is computed as $\hat{\mathbf{v}}_j = \mathbf{w}_j^H \mathbf{M}/D$. The pooled residual metric is defined as

$$L = \frac{\sum_i \|\mathbf{r}_i\|^2}{\sum_i \|\mathbf{v}_i\|^2}, \tag{2}$$

where $\mathbf{r}_i$ is the post-erase query readout residual.

3 C. Formal operator definitions

3.1 C.1 Key drift reread erase

Memory items are stored unperturbed. At erase time, local oscillator drift alters the erase key to $\tilde{\mathbf{w}}_j = \mathbf{w}_j e^{i\delta}$. The erase operator re-reads using $\tilde{\mathbf{w}}_j$ and subtracts the outer product:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D}, \qquad \mathbf{M}' = \mathbf{M} - \tilde{\mathbf{w}}_j \hat{\mathbf{v}}_j. \tag{3}$$

Because both the projection estimate and the subtractive unbinding use $\tilde{\mathbf{w}}_j$, the operation represents an exact orthogonal projection onto the subspace orthogonal to $\tilde{\mathbf{w}}_j$.

3.2 C.2 State drift reread erase

We use state drift to denote a write-time phase offset attached to the target association, followed by a mismatch between the stored phase coordinate used for re-reading and the reference coordinate used for unbinding. Specifically, the target association is stored as $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$. At erase time, the erase operator re-reads using the item's stored key coordinate $\tilde{\mathbf{w}}_j = e^{i\delta}\mathbf{w}_j$ to recover $\hat{\mathbf{v}}_j = \tilde{\mathbf{w}}_j^H \mathbf{M}/D$, but subtracts using the current reference clock key $\mathbf{w}_j$:

$$\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \hat{\mathbf{v}}_j. \tag{4}$$

3.3 C.3 Real-truncated readout channel

When evaluating the real channel, real truncation is applied to the re-read estimate before subtraction: $\hat{\mathbf{v}}_j \leftarrow \mathrm{Re}(\hat{\mathbf{v}}_j)$. At query time, the readout residual is evaluated as $\mathbf{r}_j = \mathrm{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$.

3.4 C.4 Representational erase control

The erased value is re-presented from an external target rather than re-read from the state: $\mathbf{M}' = \mathbf{M} - \mathbf{w}_j \mathbf{v}_j^\top$. This serves as a control condition separating operator-level self-estimation from state residual dynamics.

This control was executed at $D = 64$ during the O04 milestone (three seeds $\times$ six drift magnitudes, both arms; the two entries of each stored `fuga_represent` pair are two pooled events, not channels). Measured pooled residuals track the stored zero-free-parameter oracle-channel predictions within the consistency tolerance for each arm's own readout channel: maximum absolute deviation 0.020 (complex channel) and 0.017 (Re channel) over seeds and drift grid. The oracle-channel laws are insensitive to the injection variant later corrected in O04b (lab record), so the control cleanly separates self-estimation error from intrinsic post-erase residual structure.

3.5 C.5 Normalized delta control

The corrective delta-rule update modifies state $\mathbf{S} \in \mathbb{R}^{D\times D}$ via

$$\mathbf{S}_t = \mathbf{S}_{t-1} + \beta\, \mathbf{k}_t(\mathbf{v}_t^\top - \mathbf{k}_t^\top \mathbf{S}_{t-1}), \tag{5}$$

where $\mathbf{k}_t$ and $\mathbf{v}_t$ are column vectors and the outer product of $\mathbf{k}_t$ with the row residual $(\mathbf{v}_t^\top - \mathbf{k}_t^\top \mathbf{S}_{t-1})$ is taken; for complex keys and states, transposes are replaced by conjugate transposes. In the normalized-key delta control arm, keys are normalized ($\|\mathbf{k}_t\|_2^2 = 1$). For $0 < \beta < 2$, this maintains the directional feedback multiplier $|1-\beta| < 1$ strictly inside the stable unit disk.

4 D. Closed-form law derivations

Throughout this section, key entries are drawn i.i.d. with independent uniform phases over $[0, 2\pi)$. This yields $\mathbb{E}[\mathbf{w}_i^H \mathbf{w}_k] = 0$ and $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k|^2] = D$ for $i \neq k$, vanishing mixed cross-moments ($\mathbb{E}[\alpha_{jk}\alpha_{jl}^*] = 0$ for $k \neq l$, where $\alpha_{jk} = \mathbf{w}_j^H \mathbf{w}_k/D$), and circular crosstalk phasors ($\mathbb{E}[z^2] = 0$ and $\mathbb{E}[(\mathrm{Re}\, z)^2] = \mathbb{E}|z|^2/2$ for the Re-channel statements). Target vectors are deterministic and independent of the keys.

4.1 D.1 Distributed crosstalk scaling

For independent random unit-modulus keys $\mathbf{w}_i, \mathbf{w}_k \in \mathbb{C}^D$, the inner products satisfy $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k|^2] = D$ for $i \neq k$. After normalization by D, each interfering association contributes expected key-coefficient energy $\mathbb{E}[|\mathbf{w}_i^H \mathbf{w}_k/D|^2] = 1/D$ per unit squared norm of its target vector (absolute energy $\|\mathbf{v}_k\|^2/D$). Summing over $n-1$ interfering associations yields the crosstalk load $c = (n-1)/D$, consistent with distributed associative memory capacity [1–3].

4.2 D.2 Key drift under complex channel

The operator $\mathbf{P}_{\tilde{\mathbf{w}}_j}^{\perp} = \mathbf{I} - \tilde{\mathbf{w}}_j \tilde{\mathbf{w}}_j^H / D$ is the orthogonal projection onto the $(D-1)$-dimensional subspace orthogonal to $\tilde{\mathbf{w}}_j$ (the rank-one object is $\tilde{\mathbf{w}}_j \tilde{\mathbf{w}}_j^H / D$). When querying with reference key $\mathbf{w}_j = \tilde{\mathbf{w}}_j e^{-i\delta}$, the target component in $\mathbf{M}'$ along direction $\tilde{\mathbf{w}}_j$ is identically zero. Therefore, $L_{\text{key, complex}}(\delta) = 0$ in the algebraic model.

4.3 D.3 State drift under complex channel

With target item stored as $(e^{i\delta}\mathbf{w}_j)\mathbf{v}_j^\top$ and unbinding performed with reference key $\mathbf{w}_j$, the re-read estimate is:

$$\hat{\mathbf{v}}_j = \frac{\tilde{\mathbf{w}}_j^H \mathbf{M}}{D} = \mathbf{v}_j^\top + \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \tag{6}$$

The target phase cancels because the estimate is read with the item's own stored coordinate $\tilde{\mathbf{w}}_j$, while the crosstalk terms retain the $e^{-i\delta}$ factor. Subtracting $\mathbf{w}_j \hat{\mathbf{v}}_j$ yields the post-erase state:

$$\mathbf{M}' = (e^{i\delta} - 1)\mathbf{w}_j \mathbf{v}_j^\top + \sum_{k \neq j} \mathbf{w}_k \mathbf{v}_k^\top - \mathbf{w}_j \frac{e^{-i\delta}}{D} \sum_{k \neq j} \mathbf{w}_j^H \mathbf{w}_k \mathbf{v}_k^\top. \tag{7}$$

Reading with reference key $\mathbf{w}_j$ yields target residual $(e^{i\delta} - 1)\mathbf{v}_j^\top$ and interference residual

$$\mathbf{r}_{\text{cross}} = (1 - e^{-i\delta}) \frac{1}{D} \sum_{k \neq j} (\mathbf{w}_j^H \mathbf{w}_k) \mathbf{v}_k^\top. \tag{8}$$

Summing squared residual norms over stored items before taking expectations—the level at which the pooled estimator is defined—and using $\mathbb{E}[|\mathbf{w}_j^H \mathbf{w}_k|^2] = D$ for $k \neq j$ together with $\sum_j \sum_{k \neq j} \|\mathbf{v}_k\|^2 = (n-1) \sum_k \|\mathbf{v}_k\|^2$ gives

$$\frac{\sum_j \mathbb{E}[\|\mathbf{r}_j\|^2]}{\sum_j \|\mathbf{v}_j\|^2} = |e^{i\delta} - 1|^2 + |1 - e^{-i\delta}|^2 \frac{n-1}{D} = 2(1 - \cos\delta)(1 + c), \tag{9}$$

with no equal-norm assumption on the target vectors. (Per-item expectations would additionally require equal target norms and are not invoked.)

4.4 D.4 Real-truncated channel derivations

Applying real projection $\mathrm{Re}(\cdot)$ both at the re-read estimate $\hat{\mathbf{v}}_j \leftarrow \mathrm{Re}(\hat{\mathbf{v}}_j)$ and at the final query readout $\mathbf{r}_j = \mathrm{Re}(\mathbf{w}_j^H \mathbf{M}'/D)$ projects out the imaginary component of the phase differences.

- **Key drift (Re channel)**: Under key drift, the target contribution to the real query residual is $\sin^2 \delta \cdot \mathbf{v}_j^\top$, yielding target energy $\sin^4 \delta \|\mathbf{v}_j\|^2$. The interfering crosstalk from $n-1$ unperturbed associations integrates over independent random phases to pooled relative energy $\frac{c}{4}(1 - \cos 2\delta)$. Cross-terms between target and interference vanish in expectation due to zero-mean key correlations ($\mathbb{E}[\mathbf{w}_j^H \mathbf{w}_k] = 0$ for $k \neq j$). Summing gives:

$$L_{\text{key, Re}}(\delta) = \sin^4 \delta + \frac{c}{4}(1 - \cos 2\delta). \tag{10}$$

- **State drift (Re channel)**: The target association written with phase offset $e^{i\delta}$ yields an unbinding target residual $(\cos\delta - 1)\mathbf{v}_j^\top$, with energy $(1 - \cos\delta)^2 \|\mathbf{v}_j\|^2$. The unbinding subtracts the real part of the re-read crosstalk, leaving pooled interference energy $c(1 - \cos\delta)$. Independent key phases ensure vanishing cross-terms in expectation, yielding:

$$L_{\text{state, Re}}(\delta) = (1 - \cos\delta)^2 + c\,(1 - \cos\delta). \tag{11}$$

5 E. Write stability analysis

Along direction $\mathbf{k}_t$, projecting the state error yields the first-order scalar recursion

$$s_t = (1 - \beta \|\mathbf{k}_t\|^2) s_{t-1} + \beta \, q_t, \tag{12}$$

where q_t represents cross-talk from other stored patterns. The homogeneous dynamics are stable if and only if $|1 - \beta\|\mathbf{k}_t\|^2| < 1$, which requires $0 < \beta\|\mathbf{k}_t\|^2 < 2$.

With $\beta = 1$ and unnormalized embedding initialization ($\|\mathbf{k}_t\|^2 \approx 8$), the multiplier $|1 - 8| = 7$ compounds along each revisited key direction—the task draws from a finite key vocabulary that recurs across updates—inducing exponential growth of the state norm ($\|\mathbf{S}\| \approx 10^{11}$–$10^{12}$ within early epochs). Key normalization ensures $\|\mathbf{k}_t\|_2^2 = 1$; for $0 < \beta < 2$, this yields $|1 - \beta| < 1$ and keeps the directional dynamics inside the unit disk.

6 F. Experimental protocol (O05 milestone)

- **Dimensions**: Primary scale $D = 128$ ($n = 40$ pairs, predicted load $c = 39/128 \approx 0.3047$, five seeds) compared against $D = 64$ ($n = 20$ pairs, predicted load $c = 19/64 \approx 0.2969$, three seeds), achieving approximately matched load ($\approx 2.6\%$ difference).

- **Vocabulary**: Discrete alphabet of 105 tokens (48 key tokens, 48 value tokens, special marker tokens). The token vocabulary is used strictly to construct controlled key–value associative sequences; the evaluation is not intended as a natural-language modeling benchmark.

- **Arms**: `wave_complex`, `wave_re`, `normalized_delta`.

- **Training**: AdamW (learning rate 10^{-3}, weight decay 0.01, gradient-norm clip 1.0). Wave arms: 60 epochs over 600 training examples; `normalized_delta`: 150 epochs over 1200 examples. Model dimension 128, two layers. Exact operator definitions are in `prototypes/` in the repository.

- **Seeds and codebook**: The unit-modulus key codebook is fixed across all seeds (generation seed 0). The seed indexes parameter initialization and the dataset draws (train/validation pairs and probe events); seed-level variability therefore reflects optimization and data-sampling variability, not resampling of the random-key ensemble over which the closed-form expectations are taken.

- **Events and Units**: 320 forgetting events per arm and seed (exceeding pre-specified minimum of 300). For all inferential statistical tests, the random seed was treated as the independent experimental unit.

- **Drift grid**: $\delta \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$.

- **Ridge probe**: Linear Ridge regression ($\alpha = 0.1$, within-model split of 75% train [$n = 240$] / 25% test [$n = 80$]) mapping readout residuals $\mathbf{r}_j$ to continuous targets $\mathbf{v}_j \in \mathbb{R}^D$. Continuous predictions were decoded to the discrete value vocabulary by nearest cosine distance, with exact match recorded when the recovered token matched the true target token; chance level is $1/48 \approx 2.1\%$ over the 48-value vocabulary.

- **Selectivity**: Selectivity $= \mathrm{EM}_{\text{alive}} - \mathrm{EM}_{\text{erased}}$.

- **Equivalence test**: Two one-sided t-test (TOST at $\alpha = 0.05$, two-sided 90% CI) on retrieval selectivity between `wave_complex` and `normalized_delta` with pre-specified margin $\varepsilon = 0.02$. Differences are computed per seed (paired across arms) and tested against the margin with a Student-t interval (df $= 4$).

7 G. Numerical results summary

At $D = 128$, all 15 seed–arm combinations reached EM = 1.000; at $D = 64$, the nine seed–arm combinations trained to saturation (EM ≥ 0.99 in every recorded run). The four closed-form laws predicted empirical pooled residuals with maximum absolute deviation of 0.0049 at $D = 128$ and 0.0094 at $D = 64$, remaining below the 0.02 consistency tolerance. Dimension-matched load agreement for the three non-trivial cells showed maximum cross-scale difference 0.0064 under the stored pairing convention (below 0.011 under every aggregation convention examined), of which 0.0019 (state/complex, $\delta = 0.5$) is attributable to the residual load mismatch alone.

Per-seed dispersion of the pooled deviations is reported in Table 1: the stored maxima are not driven by a single outlier seed.

Table 1: Maximum absolute deviation |measured − predicted| per seed under the pooled estimator (recomputed from frozen artifacts). D=64 entries pool two events per seed; "max" is the stored value of Table 1 in the main text.

D	cell/channel	s_1	s_2	s_3	s_4	s_5	max
128	key/complex	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$
128	state/complex	0.0031	0.0004	0.0033	0.0049	0.0006	0.0049
128	key/Re	0.0006	0.0007	0.0018	0.0018	0.0013	0.0018
128	state/Re	0.0018	0.0006	0.0015	0.0011	0.0002	0.0018
64	key/complex	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	–	–	$< 10^{-4}$
64	state/complex	0.0094	0.0077	0.0086	–	–	0.0094
64	key/Re	0.0009	0.0016	0.0007	–	–	0.0016
64	state/Re	0.0015	0.0003	0.0015	–	–	0.0015

Because all closed-form laws are expectations over the random-key ensemble while every trained experiment shares a single fixed codebook (Supp. F), the consistency tolerance was calibrated by ensemble simulation: the algebraic $2 \times 2 \times \delta$ probe was repeated over $K = 48$ independent codebook-and-value draws on a grid spanning $c \in [0.08, 1.03]$ at $D = 64$ and $D = 128$ (`tests/codebook_ensemble_probe.py`, `tests/varL_calibration.py`). Across the grid, the ensemble standard deviation is well described by $\mathrm{sd}(L) = \hat{\kappa}(c)\, A(\delta)/D$, where $A(\delta) = 2(1 - \cos\delta)$ is the drift factor of the state/complex law and $\hat{\kappa}$ ranges from 1.9 to 3.7 over the entire grid, taking values $\hat{\kappa} \approx$ 2.4–2.6 at the operating load $c \approx 0.30$ for both dimensions and all drift magnitudes. Under this calibration, the trained-model deviations of Table 1 in the main text correspond to approximately one ensemble standard deviation (1.02σ at $D = 64$, 1.05σ at $D = 128$ for the state/complex cell at $\delta = 0.5$): the trained models behave as draws from the same ensemble as the algebraic construction. A complementary eight-codebook probe at $D = 64$, $n = 20$ (`tests/codebook_ensemble_probe.py`) reports per-codebook maximum absolute deviations between 0.004 and 0.024. The full calibration grid is shipped in `outputs/wave_mem/varL_calibration.json`.

In the key/complex condition, all 30 runs (five seeds $\times$ six drift magnitudes) produced algebraically zero and numerically null residuals below the numerical guard threshold $\varepsilon_{\text{null}} = 10^{-4}$. TOST yielded mean difference -0.0031 with 90% CI $[-0.0206, 0.0144]$; because the lower bound crosses $-\varepsilon = -0.02$ by 0.0006 under $n = 5$ seeds, the result is reported as inconclusive with respect to formal equivalence.

8 H. Metric audit

A simple mean of per-example ratios $\frac{1}{N}\sum_i(\|\mathbf{r}_i\|^2/\|\mathbf{v}_i\|^2)$ introduced severe estimation variance in trained models due to small target vector norms in the denominator. The primary metric was therefore standardized as the pooled ratio of sums $L = \sum_i \|\mathbf{r}_i\|^2 / \sum_i \|\mathbf{v}_i\|^2$, stabilizing the estimator.

When the post-erase residual is algebraically zero, computing erased-item exact match by taking an argmax over random noise is uninformative. In our evaluation protocol, residuals below $\varepsilon_{\text{null}} = 10^{-4}$ are recorded as numerically null residuals; the threshold is set well above the floating-point noise floor of the computation.

9 I. Negative results and scope boundaries

The vector-state geometry did not support the intended D-scaling law. The ideal wave erase and ideal delta erase were algebraically identical under normalized keys. The unnormalized $\beta = 1$ corrective delta baseline diverged due to unnormalized key norms. Learned codebooks were not included in the closed-form theory. These results define the scope of the present analysis.

References

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